

Report 5: Neural Network with Backpropagation

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May 21, 2026

1 Implementation

The neural network was implemented from scratch using object oriented programming.

The implementation contains:

- Neuron class
- Layer class
- NeuralNetwork class

The network uses:

- Sigmoid activation
- Feedforward
- Backpropagation
- Gradient descent

2 Feedforward

Each neuron computes:

$$z = \sum x_i w_i + b$$

Then applies sigmoid activation:

$$a = \sigma(z)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

3 Backpropagation

The output layer error is:

$$\delta = (y - \hat{y})\sigma'(\hat{y})$$

Weights are updated using gradient descent:

$$w = w + r\delta x$$

Where:

- r is the learning rate
- δ is the error term

4 XOR Experiment

Dataset:

```
0 0 -> 0
0 1 -> 1
1 0 -> 1
1 1 -> 0
```

Training result:

```
Training completed after 5953324 epochs!
Final loss: 0.001000
Threshold: 0.001
```

```
[0.0, 0.0] -> 0.017469
[0.0, 1.0] -> 0.984976
[1.0, 0.0] -> 0.984935
[1.0, 1.0] -> 0.015562
```

The neural network successfully learned the XOR function.

5 Training Result Figure

6 House Price Dataset

The neural network was also tested with the house price-size dataset from Lab-work 2.

The network was able to reduce the loss during training.

```
namle@192:~/dl2026/Lab4+5
~/dl2026/Lab4+5
Epoch 5921000, Loss = 0.001006
Epoch 5922000, Loss = 0.001006
Epoch 5923000, Loss = 0.001006
Epoch 5924000, Loss = 0.001006
Epoch 5925000, Loss = 0.001006
Epoch 5926000, Loss = 0.001005
Epoch 5927000, Loss = 0.001005
Epoch 5928000, Loss = 0.001005
Epoch 5929000, Loss = 0.001005
Epoch 5930000, Loss = 0.001005
Epoch 5931000, Loss = 0.001004
Epoch 5932000, Loss = 0.001004
Epoch 5933000, Loss = 0.001004
Epoch 5934000, Loss = 0.001004
Epoch 5935000, Loss = 0.001004
Epoch 5936000, Loss = 0.001003
Epoch 5937000, Loss = 0.001003
Epoch 5938000, Loss = 0.001003
Epoch 5939000, Loss = 0.001003
Epoch 5940000, Loss = 0.001003
Epoch 5941000, Loss = 0.001002
Epoch 5942000, Loss = 0.001002
Epoch 5943000, Loss = 0.001002
Epoch 5944000, Loss = 0.001002
Epoch 5945000, Loss = 0.001002
Epoch 5946000, Loss = 0.001001
Epoch 5947000, Loss = 0.001001
Epoch 5948000, Loss = 0.001001
Epoch 5949000, Loss = 0.001001
Epoch 5950000, Loss = 0.001001
Epoch 5951000, Loss = 0.001000
Epoch 5952000, Loss = 0.001000
Epoch 5953000, Loss = 0.001000

Training completed after 5953324 epochs!
Final loss: 0.001000
Threshold: 0.001

Final Results:
[0.0, 0.0] -> 0.017469
[0.0, 1.0] -> 0.984976
[1.0, 0.0] -> 0.984935
[1.0, 1.0] -> 0.015562
(dl2026) namle@192:~/dl2026/Lab4+5$
```

Figure 1: Training process and final XOR results